STA 431/631 Introduction to Probability
Homework 4

Due: Saturday, October 10, 2026

Instructions: Complete the problems below. Show all your work for full credit.

Common Problems (Required for both STA 431 and STA 631)

1. Let the joint probability mass function (PMF) of X and Y be given by:

$$f(x, y) = c(x + y^2) \quad \text{for } x \in \{1, 2\} \text{ and } y \in \{1, 2, 3\}$$

   (a) Find the value of the constant c.

   (b) Find the marginal PMFs of X and Y.

   (c) Find the conditional PMF $f_{X|Y}(x|2)$.

   (d) Are X and Y independent? Justify your answer.

2. A joint probability density function (PDF) is defined by:

$$f(x, y) = \begin{cases} c(2x + y) & \text{for } 0 < x < 1 \text{ and } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the value of c.

   (b) Find the marginal PDF of Y.

   (c) Find the conditional PDF $f_{X|Y}(x|y)$.

3. A device contains two components. The device fails if *either* component fails. The joint PDF of the lifetimes of the components X and Y, measured in hours, is given by:

$$f(x, y) = \begin{cases} \frac{x+y}{8} & \text{for } 0 < x < 2 \text{ and } 0 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

   What is the probability that the device fails during its first hour of operation?

STA 431 Only Problems

4. Let the joint PMF of X and Y be defined by:

$$f(x, y) = \begin{cases} \frac{x+y}{32} & \text{for } x \in \{1, 2\} \text{ and } y \in \{1, 2, 3, 4\} \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find $P(X > Y)$.

(b) Find $P(X + Y \leq 3)$.

5. Using the continuous joint PDF from Problem 2, $f(x, y) = \frac{1}{4}(2x + y)$ for $0 < x < 1$ and $0 < y < 2$, find the probability $P(X < Y)$.

STA 631 Only Problems

4. If the joint PDF of X and Y is given by:

$$f(x, y) = \begin{cases} 24x(1 - x - y) & \text{for } x > 0, y > 0, \text{ and } x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the marginal PDFs of X and Y.

   (b) Determine whether X and Y are independent.

5. If X is the amount of money (in dollars) that a salesperson spends on gasoline during a day and Y is the corresponding amount of money (in dollars) for which they are reimbursed, the joint PDF is given by:

$$f(x, y) = \begin{cases} \frac{1}{25}\left(\frac{20-x}{x}\right) & \text{for } 10 < x < 20 \text{ and } \frac{x}{2} < y < x \\ 0 & \text{otherwise} \end{cases}$$

   (a) Find the marginal density of X.

   (b) Find the conditional density of Y given $X = x$.

   (c) Find the probability that the salesperson will be reimbursed at least \$8 when spending \$12.